Akoizo Original Practice - 2027 Rotation Form 1

Division B | Meteorology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which satellite channel can show a jet stream in clear air?

Unit: Observing severe weather: stations, balloons, radar, and satellites.
Worked-case data: A radar image shows a storm with reflectivity of 65 dBZ, a tight red-green velocity couplet on its southwest side, and a small area of CC values around 0.7 at the same spot. What is happening?
Separate investigation observations:
Reflectivity: A hook-shaped appendix on the storm's southwest side.
Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. Visible

B. Infrared only at night

C. Radar reflectivity

D. Water vapor

Answer: _____

2. (2 points) Unsaturated air rises and cools at about:

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. 9.8 °C per km

B. 2 °C per km

C. 6.5 °C per km always

D. 20 °C per km

Answer: _____

3. (2 points) Which feature defines a supercell?

Unit: Thunderstorms and lightning.
Worked-case data: You see a flash and hear thunder 12 seconds later. How far away is the lightning, and are you safe outdoors?
Separate investigation observations:
Duration: The storm has lasted three hours.
Radar: A hook echo and a bounded weak-echo region.
Velocity: Persistent rotation in the mid levels.
Reports: Hail up to 2.75 inches (baseball size).

A. A warm front

B. A rotating updraft (mesocyclone)

C. A gust front

D. A single short-lived cell

Answer: _____

4. (2 points) A hurricane with sustained winds of 115 mph is:

Unit: Hurricanes, typhoons, and cyclones.
Worked-case data: A hurricane with 140 mph sustained winds approaches a coast with a wide, shallow continental shelf at high tide. Rate the storm and assess surge.
Separate investigation observations:
Satellite: A clear eye surrounded by a ring of very cold cloud tops.
Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

A. Category 4

B. Category 1

C. Category 3

D. Category 2

Answer: _____

5. (2 points) On Doppler velocity, green colors usually mean:

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Separate investigation observations:
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Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. Heavy rain

B. Hail

C. Motion toward the radar

D. Motion away from the radar

Answer: _____

6. (2 points) On a sounding, temperature and dew point lines close together indicate:

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. Very dry air

B. An inversion

C. Strong wind

D. Nearly saturated air

Answer: _____

7. (2 points) In the dissipating stage of a single-cell storm:

Unit: Thunderstorms and lightning.
Worked-case data: You see a flash and hear thunder 12 seconds later. How far away is the lightning, and are you safe outdoors?
Separate investigation observations:
Duration: The storm has lasted three hours.
Radar: A hook echo and a bounded weak-echo region.
Velocity: Persistent rotation in the mid levels.
Reports: Hail up to 2.75 inches (baseball size).

A. The downdraft dominates

B. The updraft strengthens

C. The cloud is still growing

D. Rain has not begun

Answer: _____

8. (2 points) Where are a hurricane’s strongest winds?

Unit: Hurricanes, typhoons, and cyclones.
Worked-case data: A hurricane with 140 mph sustained winds approaches a coast with a wide, shallow continental shelf at high tide. Rate the storm and assess surge.
Separate investigation observations:
Satellite: A clear eye surrounded by a ring of very cold cloud tops.
Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

A. Far ahead of the storm

B. In the eyewall

C. In the eye

D. In the outer rain bands

Answer: _____

9. (2 points) Why are cold, bright cloud tops on infrared images important?

Unit: Observing severe weather: stations, balloons, radar, and satellites.
Worked-case data: A radar image shows a storm with reflectivity of 65 dBZ, a tight red-green velocity couplet on its southwest side, and a small area of CC values around 0.7 at the same spot. What is happening?
Separate investigation observations:
Reflectivity: A hook-shaped appendix on the storm's southwest side.
Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. They indicate clear skies

B. They mark tall thunderstorms

C. They mark fog

D. They show surface temperature

Answer: _____

10. (2 points) An environmental lapse rate between the moist and dry rates is:

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. Absolutely unstable

B. Always neutral

C. Conditionally unstable

D. Absolutely stable

Answer: _____

11. (2 points) Interpret the adjacent opposite radial velocities cautiously.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

12. (2 points) What additional interpretation is supported by low correlation coefficient collocated with the velocity couplet?

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

13. (2 points) Calculate the surface temperature-dew-point spread.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

14. (2 points) Identify one reason that 65 dBZ reflectivity alone does not prove a tornado.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

15. (2 points) A storm moves 25 km east in 30 minutes. Find mean speed.

Storm-track table:
Time | Eastward distance
0 min | 0 km
30 min | 25 km

Answer: _____
_____
_____

16. (2 points) A rain gauge records 35 mm in 0.5 hours. Calculate mean rain rate.

Gauge table:
Elapsed time | Accumulated rain
0 h | 0 mm
0.5 h | 35 mm

Answer: _____
_____
_____

17. (2 points) A satellite IR time series shows cloud-top brightness temperature falling from 240 K to 220 K. Give a physically plausible interpretation and a limitation.

IR table:
Frame 1 | 240 K
Frame 2 | 220 K

Answer: _____
_____
_____

18. (2 points) Station wind shifts form a coherent boundary while pressure rises behind it. Describe a candidate thunderstorm-related mechanism.

Station table:
Ahead of boundary | Original wind, lower pressure
Behind boundary | Shifted wind, rising pressure

Answer: _____
_____
_____

19. (2 points) Radar detects strong winds perpendicular to its beam but a small radial velocity. Explain why the wind need not be weak.

Radar geometry table:
Wind direction | Perpendicular to beam
Radial velocity | Small
Independent wind observation | Strong

Answer: _____
_____
_____

20. (2 points) A short intense rain period falls on an already saturated basin. Explain why rainfall amount alone does not establish flood severity.

Basin table:
Antecedent soil | Saturated
Rain | Short, intense burst
Catchment drainage/routing | Not supplied

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Read the velocity difference between the adjacent west and east cells.

Same small storm area: collocated measurements

Region	Reflectivity	Radial velocity	Correlation
West cell	65 dBZ	-30 m/s	0.70
East cell	60 dBZ	+35 m/s	0.72
Background	25 dBZ	+5 m/s	0.98

F1. Synthetic dual-polarization radar panels (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Explain why the juxtaposed velocity signs support investigating rotation rather than concluding that both cells are calm.

Refer to F1. Synthetic dual-polarization radar panels (original synthetic teaching figure) shown with the first question in this figure group.

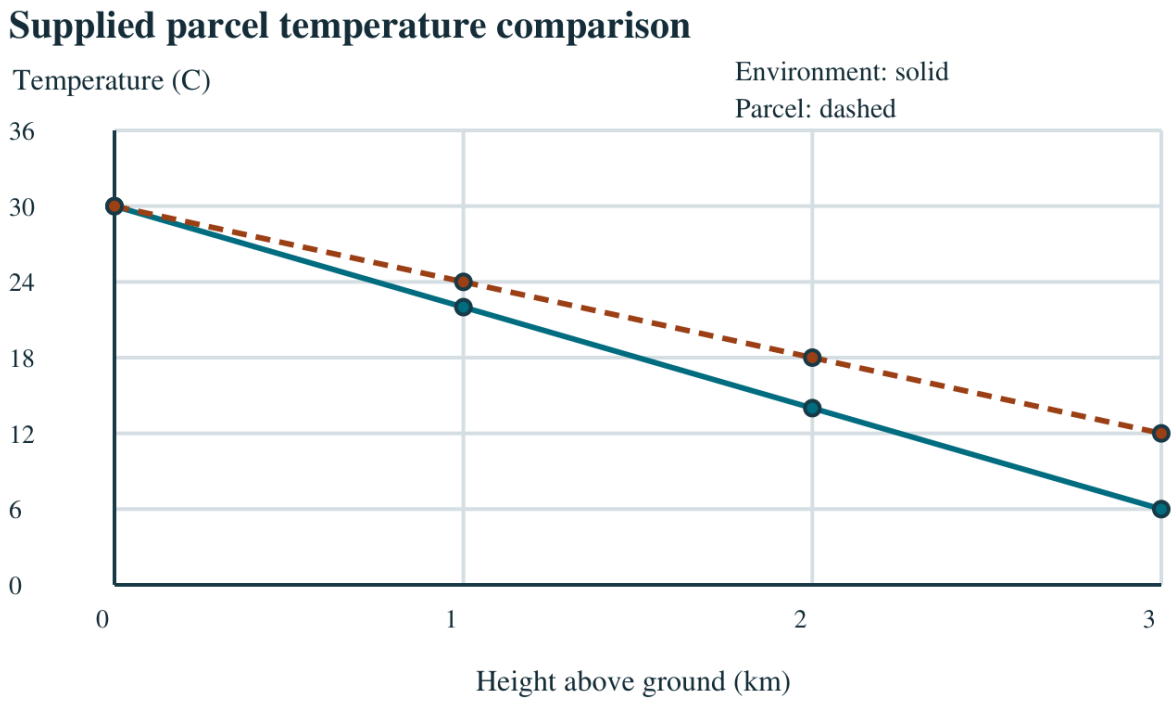
Answer: _____
_____
_____

23. (2 points) Use Figure F1. What extra interpretation does low correlation in the same area support, and what does it not prove by itself?

Refer to F1. Synthetic dual-polarization radar panels (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Read parcel-minus-environment temperature at 2 km.



F2. Parcel and environment profiles (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Under the stated simplification, is the parcel positively or negatively buoyant at 2 km? Explain.

Refer to F2. Parcel and environment profiles (original synthetic teaching figure) shown with the first question in this figure group.

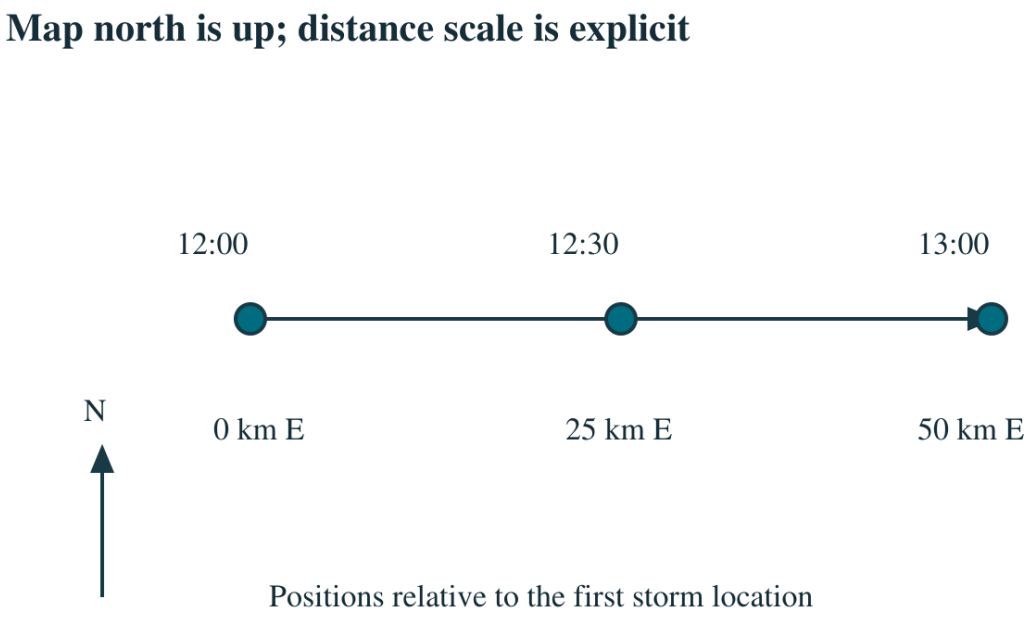
Answer: _____
_____
_____

26. (2 points) Use Figure F2. Why cannot this profile alone establish tornado occurrence?

Refer to F2. Parcel and environment profiles (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Calculate mean storm speed from 12:00 to 13:00.



F3. Storm-track map (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Give the direction of the plotted motion.

Refer to F3. Storm-track map (original synthetic teaching figure) shown with the first question in this figure group.

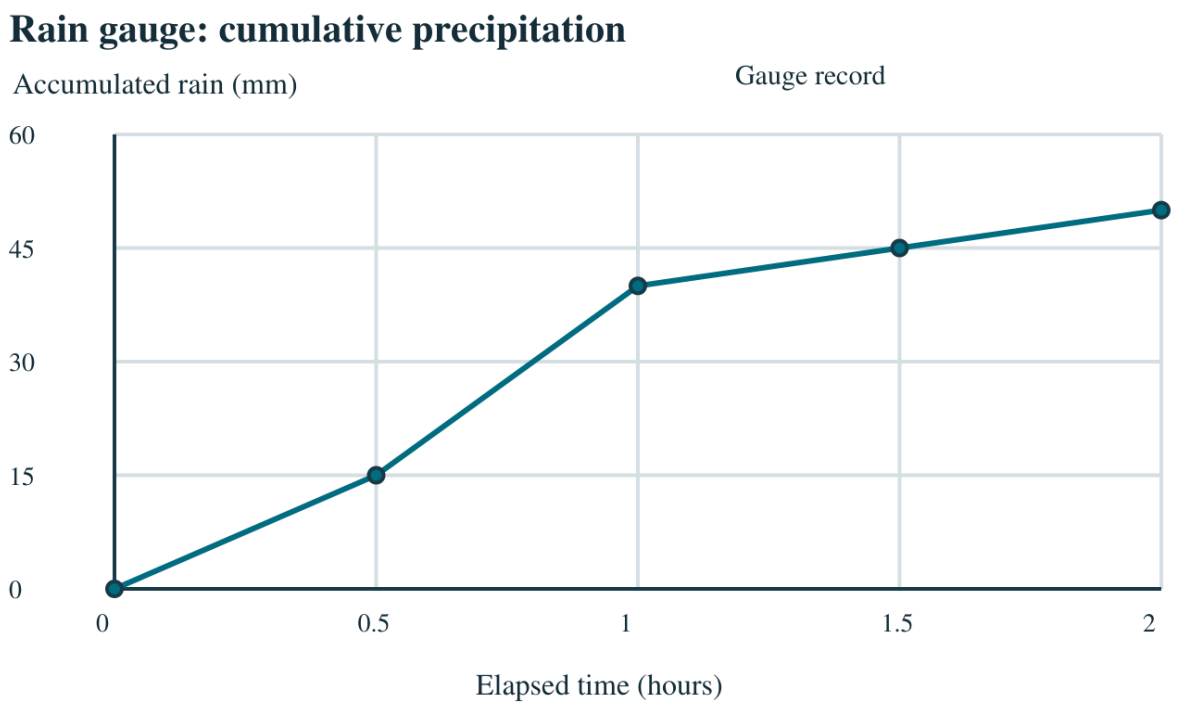
Answer: _____
_____
_____

29. (2 points) Use Figure F3. If the motion continues unchanged, how many minutes after 13:00 would a point 25 km farther east be reached?

Refer to F3. Storm-track map (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Calculate the mean rain rate during the 0.5-to-1.0 h interval.



F4. Accumulated rainfall curve (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Identify the interval with the steepest accumulation and explain what the slope measures.

Refer to F4. Accumulated rainfall curve (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. For an already saturated basin, explain why the steep interval deserves attention even if a later interval has a higher cumulative total.

Refer to F4. Accumulated rainfall curve (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____